

Homework 10

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Problem 1

Joint distributions. Random variables X, Y have joint PDF

$$f(x, y) = \begin{cases} K(x^2 + y), & 0 \leq x \leq 1, \quad 0 \leq y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find K .
- (b) Find the marginal densities f_X and f_Y , and check that each integrates to one.
- (c) Are X and Y independent?

Solution.

- (a) Because $f(x, y)$ is a PDF, it must integrate to 1 over the sample space.

$$\begin{aligned} \int_0^1 \int_0^2 f(x, y) dy dx &= \int_0^1 \int_0^2 K(x^2 + y) dy dx = 1 \\ 1 &= \int_0^1 \int_0^2 Kx^2 dy dx + \int_0^1 \int_0^2 Ky dy dx \\ 1 &= K \int_0^1 [x^2 y]_0^2 dx + K \int_0^1 \left[\frac{y^2}{2} \right]_0^2 dx \\ 1 &= K \left(\int_0^1 2x^2 dx + \int_0^1 2 dx \right) \\ 1 &= K \left(\left[\frac{2}{3} x^3 \right]_0^1 + [2x]_0^1 \right) \\ 1 &= \frac{8}{3} K \\ \boxed{K = \frac{3}{8}} \end{aligned}$$

(b)

$$f_X(x) = \int_0^2 f(x, y) dy$$

$$f_X(x) = \int_0^2 Kx^2 + Ky dy$$

$$f_X(x) = Kx^2 y \Big|_0^2 + Ky^2/2 \Big|_0^2$$

$$f_X(x) = \frac{3}{4}(x^2 + 1)$$

$$\int_0^1 f_X(x) = \int_0^1 \frac{3}{4}(x^2 + 1) dx$$

$$= \frac{3}{4} \left(\frac{x^3}{3} + x \right) \Big|_0^1$$

$$= \frac{3}{4} \left(\frac{1}{3} + 1 \right)$$

$$\int_0^1 f_X(x) = 1$$

$$f_Y(y) = \int_0^1 f(x, y) dx$$

$$= \int_0^1 Kx^2 + Ky dx$$

$$= K \frac{x^3}{3} \Big|_0^1 + Kyx \Big|_0^1$$

$$f_Y(y) = \frac{3}{8} \left(\frac{1}{3} + y \right)$$

$$\int_0^2 f_Y(y) = \frac{3}{8} \int_0^2 \frac{1}{3} + y dy$$

$$= \frac{3}{8} \left(\frac{1}{3} y + \frac{y^2}{2} \right) \Big|_0^2$$

$$= \frac{3}{8} \left(\frac{2}{3} + 2 \right)$$

$$= \frac{3}{8} \frac{8}{3}$$

$$\int_0^2 f_Y(y) = 1$$

(c) X and Y are independent if $f_{XY}(x, y) = f_X(x)f_Y(y)$

$$\begin{aligned} K(x^2 + y) &\neq \frac{3}{8} \left(\frac{1}{3} + y \right) \frac{3}{4} (x^2 + 1) \\ &= \left(\frac{3}{24} + \frac{3y}{8} \right) * \left(\frac{3x^2}{4} + \frac{3}{4} \right) \\ &= \frac{9x^2}{192} + \frac{9x^2y}{32} + \frac{9}{216} + \frac{9y}{32} \\ &= \frac{3}{8} \left(\frac{x^2}{8} + \frac{3x^2y}{4} + \frac{1}{9} + \frac{3y}{24} \right) \end{aligned}$$

They are not independent. We have an extra x^2y and constant term that don't go away, not to mention extra constants.

Problem 2

Moment generating function 1. A basketball player shoots free throws until they make their first basket. On each attempt the probability of making the shot is $q \in (0, 1)$, and shots are independent. Let X be the number of *misses* before the first make, so $X \in \{0, 1, 2, \dots\}$.

- Derive the moment generating function (MGF) of X and state where it is finite.
- Use the MGF to compute $E[X]$ and $\text{Var}(X)$.
- (*Memorylessness*) Fix an integer $k \geq 0$. Show that the MGF of $X - k$ conditional on $\{X \geq k\}$ equals $M_X(s)$, and state what this says about a player who has already missed k shots.
- Let T_r be the number of misses before the r th make. Use MGFs to identify the distribution of T_r , and give $E[T_r]$ and $\text{Var}(T_r)$.

Solution.

- The MGF for a discrete RV is $M_X(t) = \sum_x e^{tx} P(X = x)$. X is a Geometric random variable because it counts the number of trials (shots) until success. So:

$$P(X = x) = (1 - q)^x q$$

$$M_X(t) = \sum_{x=0}^{\infty} q (e^t(1 - q))^x$$

$$M_X(t) = \frac{q}{1 - e^t(1 - q)}$$

We can simplify because this is a geometric series where $a = q$ and $r = e^t(1 - q)$. It will converge when $|r| < 1$, and $e^t(1 - q)$ is always positive. So the MGF is finite for

$$t < \ln(1/(1 - q))$$

- $M'_X(0) = E[X]$, $M''_X(0) = E[X^2]$, and $\text{Var}(X) = E[X^2] - E[X]^2$.

$$\begin{aligned} M'_X(t) &= q \frac{d}{dt} \left(\frac{1}{1 - e^t + qe^t} \right) \\ &= q \left(\frac{e^t - qe^t}{(1 - e^t + qe^t)^2} \right) \\ &= \frac{(q^2 - q)e^t}{((q - 1)e^t + 1)^2} \\ M'_X(0) &= \frac{(q^2 - q)}{(q - 1 + 1)^2} \\ &= \frac{q(q - 1)}{q^2} \end{aligned}$$

$$E[X] = \frac{q - 1}{q}$$

$$\begin{aligned}
M_X''(t) &= \frac{d}{dt} \left(\frac{(q^2 - q)e^t}{((q - 1)e^t + 1)^2} \right) \\
&= \frac{((q - 1)e^t + 1)^2((q^2 - q)e^t) - ((q^2 - q)e^t)(2((q - 1)e^t + 1)((q - 1)e^t))}{((q - 1)e^t + 1)^4} \\
&= \frac{((q - 1)e^t + 1)[((q - 1)e^t + 1)((q^2 - q)e^t) - 2((q^2 - q)e^t)((q - 1)e^t)]}{((q - 1)e^t + 1)^4} \\
&= \frac{((q - 1)e^t + 1)((q^2 - q)e^t) - 2((q^2 - q)e^t)((q - 1)e^t)}{((q - 1)e^t + 1)^3} \\
&= \frac{((q - 1)e^t + 1)((q^2 - q)e^t) - 2((q^2 - q)e^t)((q - 1)e^t)}{((q - 1)e^t + 1)^3} \\
&= \frac{(q^2 - q)e^t [((q - 1)e^t + 1) - 2((q - 1)e^t)]}{((q - 1)e^t + 1)^3} \\
&= \frac{(q^2 - q)e^t [1 - (q - 1)e^t]}{((q - 1)e^t + 1)^3}
\end{aligned}$$

$$\begin{aligned}
M_X''(0) &= \frac{(q^2 - q)[1 - (q - 1)]}{((q - 1) + 1)^3} \\
&= \frac{(q^2 - q)(2 - q)}{q^3} \\
&= \frac{(q - 1)(2 - q)}{q^2}
\end{aligned}$$

$$\boxed{E[X^2] = \frac{(q - 1)(2 - q)}{q^2}}$$

$$\begin{aligned}
\text{Var}(X) &= \left(\frac{(q - 1)(2 - q)}{q^2} \right) - \left(\frac{q - 1}{q} \right)^2 \\
&= \frac{(q - 1)(2 - q) - (q - 1)^2}{q^2} \\
&= \frac{(q - 1)(2 - q - (q^2 - 2q + 1))}{q^2} \\
&= \frac{(q - 1)(2 - q - q^2 + 2q - 1)}{q^2} \\
&= \frac{(q - 1)(-q^2 + q + 1)}{q^2}
\end{aligned}$$

$$\boxed{\text{Var}(X) = \frac{-(q - 1)(q^2 - q + 1)}{q^2}}$$

(c) Let $k \geq 0$, $X \geq k$. We want to find the MGF for $P(X = t|X \geq k)$. We know:

$$P(X = t|X \geq k) = \frac{P(X = t) \cap P(X \geq k)}{P(X \geq k)} = \frac{P(X = t)}{P(X \geq k)}$$

The pmf for the numerator is:

$$P(X = t) = (1 - q)^t q$$

For the denominator we need a sum:

$$\begin{aligned} P(X \geq k) &= \sum_{j=k}^{\infty} (1 - q)^j q \\ &= q \sum_{j=k}^{\infty} (1 - q)^j \\ &= q(1 - q)^k \sum_{j=0}^{\infty} (1 - q)^j \\ &= \frac{q(1 - q)^k}{q} \\ &= (1 - q)^k \end{aligned}$$

Combining, we get:

$$\begin{aligned} \frac{P(X = t)}{P(X \geq k)} &= \frac{(1 - q)^t q}{(1 - q)^k} \\ \boxed{\frac{P(X = t)}{P(X \geq k)} &= q(1 - q)^{t-k}} \end{aligned}$$

To get the MGF, we take another sum:

$$\begin{aligned} M_{X-k|X \geq k}(t) &= \sum_{j=k}^{\infty} e^{t(j-k)} q(1 - q)^{j-k} \\ i = j - k &\implies \\ &= \sum_{i=0}^{\infty} e^{ti} q(1 - q)^i \\ \boxed{&= \frac{q}{1 - e^t(1 - q)} = M_X(t)} \end{aligned}$$

If a player has already missed k shots, their probability of making shots in the future still follows the same Geometric distribution: $X - k|X \geq k \sim \text{Geom}(q)$.

- (d) If X is the number of misses before the first make, T_r is the sum of r X_i 's, where X_i is the number of misses between shots $i - 1$ and i . Then $T_r = \sum_{i=1}^r X_i$. Because X_i are iid, we have:

$$\begin{aligned}
 M_{T_r}(t) &= E[e^{tT}] \\
 &= E[e^{t(X_1+X_2+\dots+X_r)}] \\
 &= E[e^{tX_1} \cdot e^{tX_2} \dots e^{tX_r}] \\
 &= \prod_{i=1}^r M_{X_i}(t)
 \end{aligned}$$

Then we have:

$$\begin{aligned}
 M_{T_r}(t) &= \prod_{i=1}^r M_{X_i}(t) \\
 &= \prod_{i=1}^r \frac{q}{1 - e^t(1 - q)} \\
 \boxed{M_{T_r}(t) &= \left(\frac{q}{1 - e^t(1 - q)} \right)^r}
 \end{aligned}$$

To find expectation (the first moment) we take a derivative and evaluate at $t = 0$:

$$\begin{aligned}
 \frac{d}{dt} M_{T_r}(t) &= \frac{d}{dt} \left(\frac{q}{1 - e^t(1 - q)} \right)^r \\
 &= r \left(\frac{q}{1 - e^t(1 - q)} \right)^{r-1} \left(\frac{(1 - e^t(1 - q))(0) - (q)(-e^t(1 - q))}{(1 - e^t(1 - q))^2} \right) \\
 &= r \left(\frac{q}{1 - e^t(1 - q)} \right)^{r-1} \left(\frac{e^t q (1 - q)}{(1 - e^t(1 - q))^2} \right) \\
 &= \frac{r q^r e^t (1 - q)}{(1 - e^t(1 - q))^{r+1}}
 \end{aligned}$$

$$\begin{aligned}
 M'_{T_r}(0) &= \frac{r q^r (1 - q)}{(1 - (1 - q))^{r+1}} \\
 &= \frac{r q^r (1 - q)}{q^{r+1}}
 \end{aligned}$$

$$\boxed{E[T_r] = \frac{r(1 - q)}{q}}$$

To find variance we also need the second moment.

$$\begin{aligned}
M''_{T_r}(t) &= \frac{d}{dt} \frac{rq^r e^t(1-q)}{(1-e^t(1-q))^{r+1}} \\
&= \frac{(1-e^t(1-q))^{r+1}(rq^r e^t(1-q)) - (rq^r e^t(1-q))((r+1)(1-e^t(1-q))^r)(-e^t(1-q))}{(1-e^t(1-q))^{2r+2}} \\
&= \frac{(1-e^t(1-q))^{r+1}(rq^r e^t(1-q)) + rq^r e^{2t}(1-q)^2((r+1)(1-e^t(1-q))^r)}{(1-e^t(1-q))^{2r+2}} \\
&= \frac{[rq^r e^t(1-q)(1-e^t(1-q))^r] [(1-e^t(1-q)) + e^t(1-q)(r+1)]}{(1-e^t(1-q))^{2r+2}} \\
&= \frac{[rq^r e^t(1-q)] [(1-e^t(1-q)) + e^t(1-q)(r+1)]}{(1-e^t(1-q))^{r+2}}
\end{aligned}$$

$$\begin{aligned}
M''_{T_r}(0) &= \frac{[rq^r(1-q)] [(1-(1-q)) + (1-q)(r+1)]}{(1-(1-q))^{r+2}} \\
&= \frac{[rq^r(1-q)] [q + (1-q)(r+1)]}{q^{r+2}} \\
&= \frac{[r(1-q)] [q + (1-q)(r+1)]}{q^2} \\
\boxed{E[T_r]^2} &= \frac{r(1-q) [q + (1-q)(r+1)]}{q^2}
\end{aligned}$$

So the Variance is:

$$\begin{aligned}
\text{Var}[T_r] &= E[T_r^2] - E[T_r]^2 \\
&= \left[\frac{r(1-q) [q + (1-q)(r+1)]}{q^2} \right] - \left[\frac{r(1-q)}{q} \right]^2 \\
&= \frac{r(1-q) [q + (1-q)(r+1)]}{q^2} - \frac{r^2(1-q)^2}{q^2} \\
&= \frac{r(1-q)[q + (1-q)(r+1)] - r^2(1-q)^2}{q^2} \\
&= \frac{r(1-q)[(q + r + 1 - qr - q) - r(1-q)]}{q^2} \\
&= \frac{r(1-q)[r + 1 - qr - r + qr]}{q^2} \\
\boxed{\text{Var}[T_r]} &= \frac{r(1-q)}{q^2}
\end{aligned}$$

T_r has the expectation and variance of a negative binomial distribution. A regular binomial distribution counts the number of successes after n trials, but a negative binomial counts the number of trials until n successes.

Problem 3

Mixing and conditioning. Draw $\theta \sim \text{Uniform}[0, 1]$. Conditional on Θ , let $X|\Theta \sim \text{Exponential}(\theta)$, i.e. with *rate* θ . Find the marginal PDF $f_X(x)$ and its support, justifying which θ contribute for a given x . Determine whether $E[X]$ is finite and explain.

Solution.

We know Θ is a Uniform RV, so its pdf is:

$$f_{\Theta}(\theta) = \begin{cases} 1 & 0 \leq \theta \leq 1 \\ 0 & \text{else} \end{cases}$$

Conditioning X on Θ being plugged into an Exponential means their conditional pdf is exponential:

$$f_{X|\Theta} = \begin{cases} \theta e^{-\theta x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

This first formula is from A First Course In Probability by Sheldon Ross, and we can adjust it to find $f_{X,\Theta}(x, \theta)$:

$$\begin{aligned} f_{X|\Theta}(x|\theta) &= \frac{f_{X,\Theta}(x, \theta)}{f_{\Theta}(\theta)} \\ f_{X,\Theta}(x, \theta) &= f_{X|\Theta}(x|\theta) \cdot f_{\Theta}(\theta) \quad \text{So:} \end{aligned}$$

$$f_{X,\Theta}(x, \theta) = \begin{cases} \theta e^{-\theta x} & x \geq 0, \quad 0 \leq \theta \leq 1, \\ 0 & \text{otherwise} \end{cases}$$

Assuming $x \geq 0$, we find $f_X(x)$ by integrating out the theta. Here we see which theta contribute to which x. When x is small, $e^{-\theta x} \simeq 1$, so the factor of θ in front dominates, and larger theta matter more. But when x is large, $e^{-\theta x}$ is the dominant term, so only small θ end up contributing. Recall that in this integral x is a constant so we can pull it out.

$$\begin{aligned} f_X(x) &= \int_0^1 \theta e^{-\theta x} d\theta \\ u &= \theta x, \quad du = x d\theta, \quad d\theta = \frac{1}{x} du \\ u &= (0)x = 0, \quad u = x \\ &= \int_0^x \left(-\frac{u}{x}\right) e^{-u} \frac{1}{x} du \\ &= \frac{1}{x^2} \int_0^x u e^{-u} du, \quad (\text{IBP}) \\ &= \frac{1}{x^2} \left(-u e^{-u} \Big|_0^x - e^{-u} \Big|_0^x\right) \end{aligned}$$

(Continued)

$$\begin{aligned}
&= \frac{1}{x^2} (-xe^{-x} - (e^{-x} - 1)) \\
&= \frac{1}{x^2} (-e^{-x}(x+1) + 1) \\
f_X(x) &= \begin{cases} \frac{1-(x+1)e^{-x}}{x^2} & x \geq 0 \\ 0 & x < 0 \end{cases}
\end{aligned}$$

Wait! There is an x^2 in the denominator, we need to check what happens at zero:

$$\begin{aligned}
\lim_{x \rightarrow 0^+} \frac{1 - (x+1)e^{-x}}{x^2} &\stackrel{L.H.}{=} \lim_{x \rightarrow 0^+} \frac{xe^{-x}}{2x} \\
&= \lim_{x \rightarrow 0^+} \frac{1}{2e^x} \\
&= \frac{1}{2}
\end{aligned}$$

So the pdf of X , accounting for $x = 0$, is

$$f_X(x) = \begin{cases} \frac{1-(x+1)e^{-x}}{x^2} & x > 0 \\ \frac{1}{2} & x = 0 \\ 0 & x < 0 \end{cases}$$

The region of support is $[0, \infty)$, which makes sense because we constructed X using an exponential distribution which has the same support.

Expectation:

Assuming we are in the region of support and $0 \leq \theta \leq 1$. First let us find the conditional expectation, with the intention of using the Law of Total Expectation later.

$$\begin{aligned}
\mathbb{E}[X|\Theta = \theta] &= \int_0^\infty x f_{X|\Theta}(x|\theta) dx \\
&= \lim_{t \rightarrow \infty} \int_0^t \frac{x\theta e^{-\theta x}}{1} dx \\
&= \theta \lim_{t \rightarrow \infty} \int_0^t x e^{-\theta x} dx \\
u &= \theta x, \quad du = \theta dx, \quad x = \frac{u}{\theta} \\
&= \theta \lim_{t \rightarrow \infty} \int_0^{\theta t} \frac{u}{\theta} e^{-u} \frac{du}{\theta} \\
&= \frac{1}{\theta} \lim_{t \rightarrow \infty} \int_0^{\theta t} u e^{-u} du, \quad (\text{IBP})
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\theta} \lim_{t \rightarrow \infty} [-e^{-u}(u+1)]_0^{\theta t} \\
&= \frac{1}{\theta} (1 + \lim_{t \rightarrow \infty} -e^{-\theta t}(\theta t + 1)) \\
&= \frac{1}{\theta} (1 - \lim_{t \rightarrow \infty} \frac{\theta t + 1}{e^{\theta t}}) \rightarrow \text{L.H.} \\
\mathbb{E}[X|\Theta = \theta] &= \frac{1}{\theta}
\end{aligned}$$

Now $\mathbb{E}[X]$ is $\mathbb{E}[\mathbb{E}[X|\Theta]]$, which is the Law of Total Expectation/tower property: it turns out $\mathbb{E}[X|\Theta]$ is a random variable itself, and it equals $\mathbb{E}[X|\Theta = \theta]$ when $\Theta = \theta$. So we can reasonably find its expectation by integrating over the domain of Θ . This reminds me of finding the marginal pdf by integrating out the Θ . (I read about this here, cited at bottom of problem). So because $\Theta \sim \text{Uniform}(0, 1)$:

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}\left[\frac{1}{\Theta}\right] \\
&= \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{\theta} \cdot 1 d\theta \\
&= \lim_{t \rightarrow 0^+} \ln(\theta)_t^1 \\
&= \lim_{t \rightarrow 0^+} \ln(t) \rightarrow -\infty
\end{aligned}$$

Therefore, the expectation of X is not finite.

Taboga, Marco (2021). "Conditional expectation", Lectures on probability theory and mathematical statistics. Kindle Direct Publishing. Online appendix. [link](#).

Problem 4

Moment generating function 2 (Exponential waiting times $\rightarrow$ Gaussian). Let $X \sim \text{Exponential}(\lambda)$ with density $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$.

- (a) Derive the MGF of X and use it to compute $E[X]$ and $\text{Var}(X)$. State where the MGF is finite.
- (b) Let $S_n = X_1 + \cdots + X_n$ with X_i i.i.d $\text{Exponential}(\lambda)$. Use MGFs to identify the distribution of S_n .

Solution.

- (a) Using $\phi_x(s) = \mathbb{E}[e^{sx}]$, and because $x \geq 0$, we have:

$$\begin{aligned}\phi_x(s) &= \int_0^\infty e^{sx} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^\infty e^{x(s-\lambda)} dx \\ &= \frac{\lambda}{s-\lambda} e^{(s-\lambda)x} \Big|_0^\infty\end{aligned}$$

$$\boxed{\phi_x(s) = \frac{\lambda}{\lambda - s}}$$

For $\phi_x(s)$ to be finite, we must restrict $s \in (-\infty, \lambda)$. To find the moments, it is easiest to rewrite $\phi_x(s)$ as a series as we did in class:

$$\begin{aligned}\phi_x(s) &= \frac{1}{1 - \frac{s}{\lambda}} \\ &= \sum_{n=0}^\infty \frac{s^n \mathbb{E}[x^n]}{n!} = \sum_{n=0}^\infty \frac{s^n}{\lambda^n}\end{aligned}$$

Therefore, the moments (in particular the first and second moments) are:

$$\begin{aligned}\mathbb{E}[x^n] &= \frac{n!}{\lambda^n} \\ \boxed{\mathbb{E}[x] &= \frac{1}{\lambda}} \\ \mathbb{E}[x^2] &= \frac{2}{\lambda^2}\end{aligned}$$

So the variance is:

$$\begin{aligned}\text{Var}[X] &= \mathbb{E}[x^2] - (\mathbb{E}[x])^2 \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \\ \boxed{\text{Var}[X] &= \frac{1}{\lambda^2}}\end{aligned}$$

- (b) As with problem 2d, we have a sum of iid RV's, just exponential this time. So the Mgf of S_n will be the product of $\phi_{x_i}(t)$ for $i \in \{1, \dots, n\}$. Switching parameter to t to avoid confusion.

$$\begin{aligned}\phi_{S_n}(t) &= \prod_{i=1}^n \phi_{x_i}(t) \\ &= \prod_{i=1}^n \left(\frac{1}{1 - \frac{t}{\lambda}} \right)\end{aligned}$$

$$\boxed{\phi_{S_n}(t) = \left(\frac{1}{1 - \frac{t}{\lambda}} \right)^n}$$

Let's find the expectation and variance to get more info about the distribution of S_n :

$$\begin{aligned}\phi'_{S_n}(t) &= (n) \left(\frac{\lambda}{\lambda - t} \right)^{n-1} \frac{\lambda}{(\lambda - t)^2} \\ &= \frac{n\lambda^n}{(\lambda - t)^{n+1}}\end{aligned}$$

$$\phi'_{S_n}(0) = \frac{n\lambda^n}{\lambda^{n+1}}$$

$$\boxed{\mathbb{E}[S_n] = \frac{n}{\lambda}}$$

$$\begin{aligned}\phi''_{S_n}(t) &= \frac{(\lambda - t)^{n+1}(0) - (n\lambda^n)(-(n+1)(\lambda - t)^n)}{(\lambda - t)^{2n+2}} \\ &= \frac{n\lambda^n(n+1)(\lambda - t)^n}{(\lambda - t)^{2n+2}}\end{aligned}$$

$$\phi''_{S_n}(t) = \frac{n(n+1)\lambda^n}{(\lambda - t)^{n+2}}$$

$$\phi''_{S_n}(0) = \frac{n(n+1)\lambda^n}{\lambda^{n+2}}$$

$$\boxed{\mathbb{E}[S_n^2] = \frac{n(n+1)}{\lambda^2}}$$

$$\begin{aligned}\text{Var}[X] &= \mathbb{E}[S_n^2] - \mathbb{E}[S_n]^2 \\ &= \frac{n(n+1)}{\lambda^2} - \frac{n^2}{\lambda^2} \\ &= \frac{n^2 + n - n^2}{\lambda^2}\end{aligned}$$

$$\boxed{\text{Var}[S_n] = \frac{n}{\lambda^2}}$$

The MGF, expectation, and variance are consistent with a Gamma random variable. So S_n - Gamma(n, λ).

Problem 5

Luck, Skill, and Upsets. Competitors i and j have skill scores s_i and s_j , and only the difference $s = s_i - s_j$ matters. With probability $\alpha \in [0, 1]$ a match is a pure coin flip, won by either player with probability $\frac{1}{2}$. Otherwise it is decided by skill, and i wins with the logistic probability

$$\alpha(\beta s) = \frac{1}{1 + e^{-\beta s}}, \quad \beta > 0.$$

Call α the *luck* and β the *depth of competition*.

- (a) Show that the probability that i beats j is

$$f(s) = \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{-\beta s}}.$$

Verify $f(-s) = 1 - f(s)$, give $\lim_{s \rightarrow \pm\infty} f(s)$, and interpret the limits.

- (b) Take $\alpha = 0.2$, $\beta = 1$, and $s = 2$, so i is the stronger player. Compute $P(i \text{ wins})$ and the expected net return on a \$1 wager on i at even odds. Also find the probability the coin decided the match, *given* that j won.
- (c) Let E be "the weaker player won" and F be "the match was decided by the coin," still with $s > 0$. Are E and F independent? Determine every (α, s) for which they are.
- (d) (*Why luck and depth are hard to separate.*) Expand f about $s = 0$ and show that

$$f(s) = \frac{1}{2} + \frac{(1 - \alpha)\beta}{4} s + O(s^3).$$

Explain why a dataset consisting of only closely matched contestants cannot distinguish a large luck parameter from a small depth of competition, and describe what kind of match would break the tie.

Solution.

- (a) Let C be the random variable representing who wins via the coin flip. Let L be the random variable representing who wins via the logistic probability. For both, let 1 be i wins, and 0 be j wins. Let A be the random variable representing whether we choose to use C or L . Then we can write down all our known information:

$$\begin{aligned} P(A = C) &= \alpha \\ P(A = L) &= 1 - \alpha \\ P(C = 1 | A = C) &= 0.5 = P(C = 0 | A = C) \\ P(W = 1 | A = W) &= \frac{1}{1 + e^{-\beta s}} \\ P(W = 0 | A = W) &= 1 - \frac{1}{1 + e^{-\beta s}} \end{aligned}$$

To represent the probability of i winning, we need to add up the cases above where i wins. But that's not all – we need to weight that sum by the probability that C or W is chosen. So:

$$P(i \text{ wins}) = P(A = C)P(C = 1|A = C) + P(A = W)P(W = 1|A = W)$$

$$\boxed{f(s) = \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{-\beta s}}}$$

Now let's check $f(-s) = 1 - f(s)$:

$$\begin{aligned} f(-s) &= \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{\beta s}} \\ &= \frac{\alpha}{2} + (1 - \alpha) \frac{\frac{1}{e^{\beta s}}}{1 + \frac{1}{e^{\beta s}}} \\ &= \frac{\alpha}{2} + (1 - \alpha) \frac{e^{-\beta s}}{1 + e^{-\beta s}} \\ &= \frac{\alpha}{2} + (1 - \alpha) \frac{e^{-\beta s} + 1 - 1}{1 + e^{-\beta s}} \\ &= \frac{\alpha}{2} + (1 - \alpha) \left(\frac{1 + e^{-\beta s}}{1 + e^{-\beta s}} - \frac{1}{1 + e^{-\beta s}} \right) \\ &= \frac{\alpha}{2} + (1 - \alpha) \left(1 - \frac{1}{1 + e^{-\beta s}} \right) \\ &= \frac{\alpha}{2} + (1 - \alpha) - (1 - \alpha) \frac{1}{1 + e^{-\beta s}} \\ &= 1 + \frac{\alpha}{2} - \alpha - (1 - \alpha) \frac{1}{1 + e^{-\beta s}} \\ &= 1 - \frac{\alpha}{2} - (1 - \alpha) \frac{1}{1 + e^{-\beta s}} \end{aligned}$$

$$\boxed{f(-s) = 1 - f(s)}$$

Now for the limits:

$$\begin{aligned} \lim_{s \rightarrow +\infty} f(s) &= \lim_{s \rightarrow \infty} \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{-\beta s}} \\ &= \frac{\alpha}{2} + (1 - \alpha) \lim_{s \rightarrow \infty} \frac{1}{1 + \frac{1}{e^{\beta s}}} \\ &= \frac{\alpha}{2} + (1 - \alpha) \end{aligned}$$

$$\boxed{\lim_{s \rightarrow +\infty} f(s) = 1 - \frac{\alpha}{2}}$$

If s is huge, player i is way more skilled than player j because the "skill" term $\frac{1}{1 + e^{-\beta s}}$ goes to 1. But even if i is infinitely more skilled than j , there is still the luck term that

decreases their probability of winning.

$$\begin{aligned}
 \lim_{s \rightarrow -\infty} f(s) &= \frac{\alpha}{2} + (1 - \alpha) \lim_{s \rightarrow -\infty} \frac{1}{1 + e^{-\beta s}} \\
 &= \frac{\alpha}{2} + (1 - \alpha) \lim_{s \rightarrow +\infty} \frac{1}{1 + e^{\beta s}} \\
 &= \frac{\alpha}{2} \\
 \boxed{\lim_{s \rightarrow -\infty} f(s) = \frac{\alpha}{2}}
 \end{aligned}$$

This is the opposite situation, where player j is infinitely more skilled than player i . Even if player i is infinitely bad, they are still going to win the coin flip matches half the time.

(b) With $\alpha = 0.2$, $\beta = 1$, $s = 2$, we have:

$$\begin{aligned}
 f(2) &= \frac{0.2}{2} + 0.8 \frac{1}{1 + e^{-2}} \\
 &\approx 0.8046
 \end{aligned}$$

So the probability i wins is about 80%.

Computations for the next parts were done in Desmos. If we make a \$1 wager on i at even odds and i wins, we make \$1, and if i loses, we make -\$1. So the expected net return R is:

$$R = P(i \text{ wins}) - P(i \text{ loses})$$

$$\boxed{R \approx 0.6093}$$

The probability the coin decided the match is $P(A = C)$. The probability that j wins is $1 - f(x)$, so in this case $P(j \text{ wins}) = 0.1953$. We need to find $P(A = C | j \text{ wins})$.

$$\begin{aligned}
 P(A = C | j \text{ wins}) &= \frac{P(A = C \& j \text{ wins})}{P(j \text{ wins})} \\
 &= \frac{\frac{0.2}{2}}{0.1953}
 \end{aligned}$$

$$\boxed{P(A = C | j \text{ wins}) = 0.5119}$$

(c) Let E is the RV representing 'the weaker player won' and F is the RV representing 'the match was decided by the coin' ($F = A$ from above). E, F are independent only when

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = P(E)$$

So we need to find some probabilities. The players being perfectly matched in skill is $s = 0$. So as long as $s > 0$, the weaker player will be j .

$$P(E) = 1 - \frac{\alpha}{2} - (1 - \alpha) \frac{1}{1 + e^{-\beta s}}$$

The match being decided by the coin is $P(F) = \alpha$. The probability the weaker player won *and* the match was decided by the coin is $\alpha/2$. So we have:

$$\begin{aligned} P(E|F) &= \frac{\frac{\alpha}{2}}{\alpha} \\ &= \frac{1}{2} \end{aligned}$$

Makes sense. But can this ever be $P(E)$? We know as $s \rightarrow +\infty$, $f(s) = 1 - \frac{\alpha}{2}$. So:

$$\begin{aligned} \lim_{s \rightarrow +\infty} P(E) &= 1 - 1 + \frac{\alpha}{2} \\ &= \frac{\alpha}{2} \end{aligned}$$

Then we would need $\alpha = 1$, which just means the match would always be decided by the coin: so at $(+\infty, \alpha)$, E and F are independent. Otherwise, if skill is involved, whether the weaker player wins is very affected by whether the match was decided by the coin.

(d) We want to expand f about $s = 0$. Let's take derivatives!

$$\begin{aligned} f(s) &= \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{-\beta s}} \\ f(0) &= \frac{\alpha}{2} + (1 - \alpha) \frac{1}{2} = \frac{1}{2} \\ f'(s) &= (1 - \alpha) \frac{-1(-\beta e^{-\beta s})}{(1 + e^{-\beta s})^2} \\ &= (1 - \alpha) \frac{\beta e^{-\beta s}}{(1 + e^{-\beta s})^2} \\ f'(0) &= \frac{(1 - \alpha)\beta}{4} \\ f''(s) &= (1 - \alpha) \frac{(1 + e^{-\beta s})^2(-\beta^2 e^{-\beta s}) - (\beta e^{-\beta s})(2(1 + e^{-\beta s})(-\beta e^{-\beta s}))}{(1 + e^{-\beta s})^4} \\ &= (1 - \alpha) \frac{-\beta^2 e^{-\beta s}(1 + e^{-\beta s})^2 + 2\beta^2 e^{-2\beta s}(1 + e^{-\beta s})}{(1 + e^{-\beta s})^4} \\ &= (1 - \alpha) \frac{(1 + e^{-\beta s}) [-\beta^2 e^{-\beta s}(1 + e^{-\beta s}) + 2\beta^2 e^{-2\beta s}]}{(1 + e^{-\beta s})^4} \\ &= \beta^2(1 - \alpha)e^{-\beta s} \cdot \frac{2e^{-\beta s} - (1 + e^{-\beta s})}{(1 + e^{-\beta s})^3} \\ &= \frac{2\beta^2(1 - \alpha)e^{-2\beta s}}{(1 + e^{-\beta s})^3} - \frac{\beta^2(1 - \alpha)e^{-\beta s}}{(1 + e^{-\beta s})^2} \end{aligned}$$

$$\begin{aligned}
f''(0) &= \frac{2\beta^2(1-\alpha)}{(2)^3} - \frac{\beta^2(1-\alpha)}{(2)^2} \\
&= \frac{\beta^2(1-\alpha) - \beta^2(1-\alpha)}{4}
\end{aligned}$$

$$\boxed{f''(0) = 0}$$

Then the expansion of f about $s = 0$ is

$$f(s) = \frac{1}{2} + \frac{(1-\alpha)\beta}{4}s + \mathcal{O}(s^3)$$

Now. If a dataset only has closely matched contestants, that means s will be fairly small. If α is large, $(1-\alpha)$ is small, which results in a small second term in the expansion. If β is small, that also results in a small second term. If they are both large, the term is large. Since $(1-\alpha)$ and β have the same effect on the most dominant non-constant term, we can't really distinguish their effects.

To break the tie, we need a larger s . As shown in the limits in part (a), for large skill difference, the win probability for each player is only determined by alpha.

Problem 6

Simulation Assignment 1: Coupled Inequalities. Let X, Y, Z be independent, Uniform $[0,1]$ random variables.

- (a) Analytically compute the probability that

$$X^2 + Y^2 < Z \quad \text{and} \quad Z^2 > XY.$$

Describe the region of the unit cube satisfying both inequalities and set up the integrals. (Hint: polar coordinates in x, y . The two lower bounds on z exchange dominance at some $r_*(\theta)$).

- (b) Estimate the same probability by Monte Carlo and compare. Plot the estimate against the sample size N with a logarithmic horizontal axis, and draw the analytic value as a horizontal line.

Solution.

- (a) The two lower bounds are $x^2 + y^2$ and xy . We want a total lower bound of $\max_{x,y}\{x^2 + y^2, xy\}$. To figure out where those 3d objects exchange dominance, it's best to use polar coordinates. So armed with our trusty:

$$\begin{aligned} x^2 + y^2 &= r^2 \\ x &= r \cos(\theta) \\ y &= r \sin(\theta) \\ \sqrt{xy} &= r \sqrt{\cos(\theta) \sin(\theta)} \end{aligned}$$

$r_*(\theta)$ is where the two are equivalent:

$$\begin{aligned} r^2 &= r \sqrt{\cos(\theta) \sin(\theta)} \\ \boxed{r_*(\theta) &= \sqrt{\cos(\theta) \sin(\theta)}} \end{aligned}$$

So we know:

$$z > \begin{cases} r \sqrt{\cos(\theta) \sin(\theta)}, & r < r_*(\theta) \\ r^2, & r > r_*(\theta) \end{cases}$$

Now we can set up the integration using cylindrical coordinates. Note $\theta \in [0, \frac{\pi}{2}]$ because we are in the positive quadrant.

$$\begin{aligned}
P &= \int_0^{\pi/2} \int_0^{\sqrt{\cos(\theta) \sin(\theta)}} \int_{r\sqrt{\cos(\theta) \sin(\theta)}}^1 r dz dr d\theta + \int_0^{\pi/2} \int_{\sqrt{\cos(\theta) \sin(\theta)}}^1 \int_{r^2}^1 r dz dr d\theta \\
&= \int_0^{\pi/2} \int_0^{\sqrt{\cos(\theta) \sin(\theta)}} r \left(1 - r\sqrt{\cos(\theta) \sin(\theta)} \right) dr d\theta + \int_0^{\pi/2} \int_{\sqrt{\cos(\theta) \sin(\theta)}}^1 r (1 - r^2) dr d\theta \\
&= \int_0^{\pi/2} \left(\frac{r^2}{2} - \frac{r^3}{3} \sqrt{\cos(\theta) \sin(\theta)} \right) \Big|_0^{\sqrt{\cos(\theta) \sin(\theta)}} d\theta + \int_0^{\pi/2} \left(\frac{r^2}{2} - \frac{r^4}{4} \right) \Big|_{\sqrt{\cos(\theta) \sin(\theta)}}^1 d\theta \\
&= \int_0^{\pi/2} \frac{\cos(\theta) \sin(\theta)}{2} - \frac{(\cos(\theta) \sin(\theta))^2}{3} d\theta + \int_0^{\pi/2} \frac{1}{4} - \frac{\cos(\theta) \sin(\theta)}{2} + \frac{(\cos(\theta) \sin(\theta))^2}{4} d\theta \\
&= \int_0^{\pi/2} \frac{\cos(\theta) \sin(\theta)}{2} - \frac{(\cos(\theta) \sin(\theta))^2}{3} + \frac{1}{4} - \frac{\cos(\theta) \sin(\theta)}{2} + \frac{(\cos(\theta) \sin(\theta))^2}{4} d\theta \\
&= \int_0^{\pi/2} -\frac{(\cos(\theta) \sin(\theta))^2}{12} + \frac{1}{4} d\theta \\
&= \frac{\pi}{8} - \frac{1}{12} \int_0^{\pi/2} (\cos(\theta) \sin(\theta))^2 d\theta \\
&= \frac{\pi}{8} - \frac{1}{12} \int_0^{\pi/2} \left(\frac{\sin(2\theta)}{2} \right)^2 d\theta \\
&= \frac{\pi}{8} - \frac{1}{48} \int_0^{\pi/2} \sin^2(2\theta) d\theta \\
&= \frac{\pi}{8} - \frac{2\pi}{192} \\
\boxed{P = \frac{23\pi}{192}}
\end{aligned}$$

- (b) Tested for two different N's. In both cases the convergence is very wiggly, but I think the shape is more interesting with more samples.

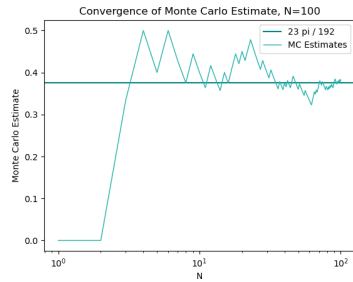


Figure 1

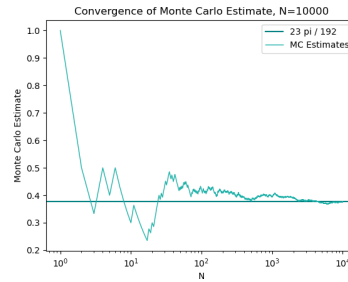


Figure 2

Code

```
1  ## IMPORTS #####
2  #####
3  import numpy as np
4  import matplotlib.pyplot as plt
5
6  ## Monte Carlo #####
7  #####
8  def monte_carlo(X, Y, Z):
9      """
10     does things frfr
11     Inputs:  X, Y, Z....(ints) sampled from U ~(0,1)
12     Outputs: 1 if point accepted, 0 else.
13     """
14     bound1 = X**2+Y**2
15     bound2 = X*Y
16
17     if bound1 < Z and bound2 < Z**2:
18         return 1
19     else:
20         return 0
21
22
23  ## Main #####
24  #####
25  if __name__ == "__main__":
26      N = 100
27      count = 0
28      estimates = []
29
30      for i in range(1, N+1):
31          X, Y, Z = np.random.rand(3)
32          count += monte_carlo(X,Y,Z)
33          estimates.append(count/i)
34          i += 1
35
36      plt.axhline(y=(23*np.pi)/192, color = "teal", label = "23 pi / 192")
37      plt.plot(range(1, N+1), estimates, color = "lightseagreen", linewidth
38              = 1, label = "MC Estimates")
39      plt.xscale("log")
40      plt.xlabel("N")
41      plt.ylabel("Monte Carlo Estimate")
42      plt.title(f"Convergence of Monte Carlo Estimate, N={N}")
43      plt.legend()
44      plt.savefig(f"HW1_Monte Carlo_{N}.png")
```